THE UNIVERSAL PROPERTY OF THE FLATTENING IDEAL OVER AN ARTINIAN RING

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Existence of the smallest ideal modulo which the module becomes flat over an Artinian ring. Lemma \ref{lemma-flattening-artinian}, cited at proof lines 4037-4038. Provides the ideal whose universal property is proved.
- Flatness is preserved by base change. Algebra, Lemma \ref{algebra-lemma-flat-base-change}, cited at proof lines 4039-4040. Gives one implication.
- Flatness descends along an injective map of Artinian rings. Algebra, Lemma \ref{algebra-lemma-descent-flatness-injective-map-artinian-rings}, cited at proof lines 4043-4045. Yields flatness of the module modulo the kernel of the base change.

Source reference. Original lemma 0525: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6558–6558. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 0524: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6557–6557. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 00HI: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 644–644. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

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Source reference. Original lemma 051L: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6539–6539. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let R be an Artinian ring. Let M be an R-module. Let $I \subset R$ be the smallest ideal $I \subset R$ such that M/IM is flat over R/I. Then I has the following universal property: For every ring map $\varphi : R \to R'$ we have*

$$R' \otimes_R M \text{ is flat over } R' \Leftrightarrow \text{we have } \varphi(I) = 0.$$

Proof. Note that I exists by Lemma 0524. The implication $\Rightarrow$ follows from Algebra, Lemma 00HI. Let $\varphi : R \to R'$ be such that $M \otimes_R R'$ is flat over R'. Let $J = \mathrm{Ker}(\varphi)$. By Algebra, Lemma 051L and as $R' \otimes_R M = R' \otimes_{R/J} M/JM$ is flat over R' we conclude that M/JM is flat over R/J. Hence $I \subset J$ as desired. □